

Multi-UPF Traffic Steering for 5G Network Slices in Stadium and Mass Event Deployments

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Abstract—This abstract briefly states the problem, the proposed approach, key methods, and the main results in about 150–250 words, following IEEE style.

Index Terms—5G network slicing, User Plane Function (UPF) selection, multi-access edge computing (MEC), quality of service (QoS), radio access network (RAN)

ABBREVIATIONS

Abbreviation	Full Form
5G	Fifth Generation
AR	Augmented Reality
eMBB	Enhanced Mobile Broadband
IoT	Internet of Things
MEC	Multi-Access Edge Computing
mMTC	Massive Machine-Type Communication
NR	New Radio
QoE	Quality of Experience
QoS	Quality of Service
RAN	Radio Access Network
SLA	Service-Level Agreement
UE	User Equipment
UPF	User Plane Function
URLLC	Ultra-Reliable Low-Latency Communication
VR	Virtual Reality

I. INTRODUCTION

Stadiums and large-scale public events are increasingly becoming testbeds for next-generation connectivity. As 5G NR deployments mature, the distinct service classes they support open up possibilities that were simply not feasible with previous network generations: eMBB for high-throughput fan engagement, URLLC for mission-critical operations such as drone coordination, and mMTC for dense IoT sensing. Delivering immersive AR/VR experiences to tens of thousands of spectators simultaneously, while also supporting real-time safety systems and pervasive crowd monitoring, places extraordinary demands on network infrastructure. Latencies must remain below 20 ms for time-sensitive applications, throughput must scale to gigabit levels, and reliability cannot drop below 99.999% for safety-critical slices. Meeting all of these requirements in a single venue, at peak load, is a genuinely hard problem.

A key part of the solution lies at the edge. By deploying MEC nodes close to the RAN, operators can offload compute-intensive tasks to local servers, including 360° video processing, AI-based crowd analytics, and real-time resource orchestration. Moving this processing away from the core cuts round-trip latency significantly and improves the QoE for end users. The growing availability of localized UPFs, driven in part by the broader expansion of distributed data centers, makes this architecture increasingly practical. However, relying on a single centralized UPF in a high-density scenario quickly becomes a bottleneck: bandwidth gets congested, propagation delays creep up, and the QoS guarantees associated with each network slice begin to erode. A multi-UPF deployment addresses this by steering traffic to whichever edge node is best placed to serve a given UE, provided the selection logic is sound. Getting that logic right requires algorithms that can jointly account for load, latency, per-slice QoS requirements, and fairness across UEs, so that neither eMBB fans nor URLLC safety services are starved of resources. This paper proposes and evaluates a set of such algorithms, with the aim of enabling reliable 5G edge service delivery across all slice types in mass event environments.

II. RELATED WORK

The transition from hardware-centric legacy infrastructure to the 5G Next Generation System (NGS) is fundamentally shaped by the adoption of a Service-Based Architecture (SBA). A cornerstone of this shift is Control and User Plane Separation (CUPS), which decouples session management logic from data-plane routing, enabling the UPF to be instantiated anywhere in the topology—from centralized data centers to the network edge. This architectural flexibility transforms UPF selection from a static assignment into a dynamic, slice-aware optimization problem, and provides the architectural motivation for the work surveyed in this section. The body of work relevant to this paper spans three interconnected areas: 5G network slicing and resource management, UPF placement and selection in mobile core networks, and MEC deployment in high-density environments.

Network slicing has been extensively studied since its introduction as a core concept in 3GPP Release 15. Early work focused on the radio access side, with proposals for slice-aware scheduling and resource partitioning at the base

station level []. More recent contributions have addressed end-to-end slice management, including control-plane signaling, isolation guarantees, and dynamic admission control across eMBB, URLLC, and mMTC traffic classes []. These slices impose distinct KPI targets: eMBB demands peak data rates of 20 Gbps downlink and 10 Gbps uplink; URLLC mandates end-to-end latency below 1 ms with 99.999% availability; and mMTC targets connection densities of up to 10^6 devices/km². []. While these works establish the slicing framework, they largely assume a fixed core network topology and do not consider UPF placement as a variable in the optimization. More recently, distributed QoS-aware slicing algorithms have been extended beyond 5G; [1] proposes an ADMM-based framework combining graph neural networks with distributed optimization to achieve provably convergent resource slicing under dynamic and unreliable network conditions.

UPF placement and traffic steering have received growing attention as operators move toward distributed core architectures. Several studies have framed UPF placement as a facility location problem, optimizing for latency or load under capacity constraints [2]. Others have proposed dynamic UPF migration triggered by mobility patterns or traffic shifts. However, most of these approaches target general LTE/5G deployments and do not account for the extreme density and heterogeneous QoS profiles characteristic of mass event scenarios, where eMBB, URLLC, and mMTC traffic coexist and compete for the same edge infrastructure. An promising EGT-based approach for dynamic UPF selection, where UEs adapt their choice to minimize end-to-end delay using replicator dynamics converging to a fair equilibrium, is proposed in [3]. Complementing this, [4] proposes an ML-based framework that proactively schedules UPF placement and chaining reconfigurations by predicting latency violations, reducing unnecessary reconfiguration events by over 50% while keeping QoS within bounds more than 99.8% of the time.

MEC deployment in high-density venues has been explored in the context of sports stadiums and concerts, with work addressing task offloading, video caching, and AR/VR. Notably, Chochliouros et al. validate a two-tier MEC architecture with Cloud-Enabled Small Cells at a live football stadium, demonstrating that local edge processing effectively reduces backhaul load in high-density venues [5], [6]. streaming []. These studies demonstrate the latency and throughput benefits of edge processing but typically treat the UPF as a fixed anchor point rather than a selectable network element. The interaction between MEC service placement and UPF selection under slice-specific QoS constraints remains largely unaddressed in the literature. A further enabler of real-time UPF selection is the signaling efficiency introduced by 5G New Radio (NR). Unlike LTE, which transmits cell-specific reference signals (CRS) four times per millisecond [7], 5G NR uses a default 20 ms synchronization signal block (SSB) periodicity [8], substantially reducing air-interface overhead and permitting more accurate, low-latency state collection from candidate UPFs without saturating the channel. Additionally, 5G NR introduces flexible mini-slot scheduling of 2, 4, or 7 OFDM

symbols with variable start positions [9], [10], enabling the bursty, asynchronous transmissions typical of URLLC and dense mMTC traffic. These radio-layer properties tighten the coupling between UPF selection granularity and physical-layer scheduling, reinforcing the need for a selection framework that is both slice-aware and responsive to current node load. This paper bridges these gaps by jointly considering UPF selection and slice-aware resource allocation in a multi-UPF, multi-MEC architecture, evaluated specifically in the context of mass event deployments where traffic heterogeneity and peak-load conditions make the selection problem non-trivial.

III. METHODOLOGY

A. System Model and Network Topology

The network is modelled as a directed graph $\mathcal{G}(\mathcal{N}, \mathcal{E})$, where $\mathcal{N}$ denotes physical nodes and $\mathcal{E}$ directed links [4]. The node set comprises stadium-side gNBs forming the 5G NR RAN, a MEC edge server $c_{\text{mec}} \in \mathcal{N}_c$ at a stadium aggregation point, and a central cloud server $c_{\text{cc}} \in \mathcal{N}_c$ in the operator core. Each server c has maximum processing capacity C_c . Each link $(u, v) \in \mathcal{E}$ carries a bandwidth capacity $\beta_{u,v}$ and one-way delay $d_{u,v}$ comprising propagation and switching components.

UPF_{mec} provides low propagation latency at constrained processing capacity. UPF_{cc} provides higher processing headroom at greater propagation cost. Propagation delay is modelled at 5 $\mu\text{s}/\text{km}$ of optical fibre [3], and the central UPF is sited such that $t_{\text{prop}_{\text{cc}}}/t_{\text{prop}_{\text{mec}}} = 4$.

Per 3GPP TS 23.501 [11], UPF selection and session management are the responsibility of the SMF. In this work the EGT algorithm of Section III-F runs within the SMF as a population-aware steering policy, maintaining population state vectors $\mathbf{x}^g(t)$ and issuing anchor assignments to UEs via PFCP over the N4 reference point [12].

The UE population is partitioned into two slice groups $g \in \{1, 2\}$. Group $g = 1$ comprises eMBB users (S-NSSAI: SST=1) who represent the dominant traffic class at a stadium. eMBB traffic in this scenario primarily consists of HD video replay consumption, social media upload, and venue app usage. Active HD video streaming demands approximately 25 Mbps per device [13], while the average per-device load in a dense shared-cell environment, as observed in large-event deployments, is significantly lower due to scheduler sharing across concurrent users [14], [15]. We model $\rho_{\text{eMBB}} = 10$ Mbps as the mean offered load per eMBB UE under peak conditions, which is conservative relative to peak single-device rates but reflects realistic concurrent cell loading.

Group $g = 2$ comprises URLLC users (S-NSSAI: SST=2) supporting latency-sensitive AR/VR overlay services offloaded to the MEC node. In this deployment, URLLC sessions serve real-time AR rendering tasks in which compute-intensive frame processing is handled at the MEC server and only compressed output is delivered to headsets, requiring $\rho_{\text{URLLC}} = 25$ Mbps per session [13] with a strict packet delay budget. Both the per-slice packet delay budgets PDB_g and 5QI values are derived from Table 5.7.4-1 of 3GPP TS 23.501 [11] and TS 22.261 [16]: $\text{PDB}_1 = 300$ ms for eMBB video (5QI 6,

non-GBR) and $PDB_2 = 10$ ms for interactive AR/VR (5QI 82, Delay-Critical GBR) [16]. All parameters are listed in Table I.

TABLE I
SYSTEM, SLICE, AND ALGORITHM PARAMETERS

Symbol	Value	Description and source
<i>Network topology</i>		
C_c^{mec}	40 CPU	MEC server capacity (edge-constrained)
C_c^{cc}	120 CPU	Central server capacity
$\beta_{u,v}$	10 Gbps	Link bandwidth
$t_{\text{prop}_{\text{mec}}}$	0.25 ms	Propagation delay to local UPF
$t_{\text{prop}_{\text{cc}}}$	1.0 ms	Propagation delay to central UPF
<i>Slice parameters (3GPP TS 22.261 [16], TS 23.501 [11])</i>		
M_1	800	eMBB UE population (SST=1)
M_2	200	URLLC UE population (SST=2)
ρ_{eMBB}	10 Mbps	Mean per-UE eMBB load at peak [14], [15]
ρ_{URLLC}	25 Mbps	Per-session AR/VR offload load [13]
PDB_1	300 ms	eMBB PDB, 5QI 6 [11]
PDB_2	1020 ms	URLLC AR/VR PDB, 5QI 82 [16]
<i>UPF delay model [3]</i>		
k_{mec}	4.89	Processing exponent (capacity-normalised load), local UPF
k_{cc}	0.49	Processing exponent (capacity-normalised load), central UPF, ratio $k_{\text{mec}}/k_{\text{cc}} = 10$ per [3]
$C_{\text{mec}}^{\text{cap}}$	13,000 Mbps	MEC UPF throughput capacity
$C_{\text{cc}}^{\text{cap}}$	40,000 Mbps	Central UPF throughput capacity
<i>EGT algorithm [3]</i>		
β	1	Payoff scaling constant
α	2	Review rate upper bound
ε	0.01	Convergence threshold

B. Traffic Dynamics at Mass Events

A defining characteristic of the stadium scenario is the pronounced non-stationarity of traffic demand. Real-world measurements at large events document this pattern clearly. At Super Bowl LIII, 24 TB of data were transferred within the stadium alone, with the data network spiking to 13.06 Gbps during the halftime show from over 30,000 concurrent mobile devices [17]. At Super Bowl LVII, a peak Wi-Fi throughput of 25.8 Gbps was recorded, and the halftime show drove a 450% increase in bandwidth usage relative to the pre-show baseline [14]. At the 2023 Formula 1 Grand Prix in Mexico City, 5G accounted for 40% of over 117 TB of total mobile data traffic, with average 5G download speeds of 274 Mbps for connected users [15]. These measurements establish three distinct load phases that our model must capture.

Pre-match ramp-up (Phase I). Spectators enter the venue over 30–60 minutes. eMBB sessions increase monotonically as users connect, stream highlights, and access venue applications. Load is moderate and UPF_{mec} can absorb the majority of sessions with favourable delay; the EGT equilibrium concentrates sessions at the edge node.

Peak-load phase (Phase II). At kick-off and halftime, concurrent eMBB and URLLC sessions generate the aggregate demand surge documented in field measurements. The exponential processing delay model in Eq. (3) captures the rapid onset of queuing congestion at UPF_{mec} as aggregate load approaches its capacity C_c^{mec} . Rising processing delay reduces the payoff of the local UPF below the population average, and the replicator dynamics redistribute sessions to UPF_{cc} . Concurrently, URLLC sessions supporting AR/VR overlay rendering must maintain $PDB_2 = 10$ ms end-to-end; if congestion at the local UPF would push delay beyond this budget, those sessions are steered to the central UPF provided the total combined delay including propagation still satisfies the 5QI 82 constraint. This phase creates the conditions under which the EGT equilibrium may transiently lag behind rapid load shifts, motivating the ML reconfiguration trigger in Section V.

Post-match dispersal (Phase III). Session counts fall as spectators leave. The equilibrium reverts toward UPF_{mec} , but active URLLC sessions must not be forcibly re-anchored during dispersal, as unnecessary PDU session modifications introduce signalling overhead and momentary service interruption.

C. Justification for Evolutionary Game Theory

The UPF session-steering problem in a stadium environment has four properties that make EGT the appropriate decision framework.

Population-scale decentralisation. At peak load, up to $M_1 + M_2 = 1000$ PDU sessions must be steered concurrently. Centralised formulations such as integer linear programming (ILP) are computationally intractable at this scale and under dynamic demand [4]. EGT operates on aggregate population fractions $\mathbf{x}^g(t)$ rather than individual sessions, reducing the SMF's per-iteration complexity from $O(M_g)$ to $O(|S_g|) = O(2)$ for a two-UPF strategy space [3].

Non-stationarity. The three-phase load profile changes continuously. Unlike static assignment policies, which require offline re-optimisation to respond to load changes, the replicator dynamics adapt $\mathbf{x}^g(t)$ continuously to the prevailing payoff landscape without requiring a global re-solve.

Convergence to a fair equilibrium. The replicator dynamics converge to a Nash equilibrium at which all active strategies within each group achieve equal payoff [3]. At equilibrium, neither UPF is overloaded relative to the other given the current population mix, which enforces load-balancing fairness across the eMBB and URLLC slice groups simultaneously. Alevizaki et al. demonstrate convergence in under 100 iterations [3], making the approach feasible within 5G session management timescales.

Low N4 signalling overhead. The SMF maintains population state internally; no per-session global state exchange is required. The algorithm issues N4/PCF session modification rules only for newly established PDU sessions and reconfiguration events, consistent with the PCF interaction model of 3GPP TS 29.244 [12].

D. UPF Strategy Space and Population State

For each group $g \in \{1, 2\}$, the strategy set is $S_g = \{\text{UPF}_{\text{mec}}, \text{UPF}_{\text{cc}}\}$. The SMF maintains the population state of group g as:

$$\mathbf{x}^g(t) = [x_{\text{mec}}^g(t), x_{\text{cc}}^g(t)], \quad x_{\text{mec}}^g + x_{\text{cc}}^g = 1 \quad (1)$$

where $x_i^g(t)$ is the fraction of sessions in group g currently anchored at UPF_i , computed from per-UPF session counts reported over N4/PFCP at each iteration interval Δt .

E. Delay Model and Payoff Function

The end-to-end delay for a session in group g anchored at UPF_i is:

$$d_i^g(\mathbf{x}) = t_{\text{prop}_i}^g + t_{\text{UPF}_i}(\mathbf{x}) \quad (2)$$

The UPF processing and queuing delay is modelled as an exponential function of aggregate offered load at that node [3]:

$$t_{\text{UPF}_i}(\mathbf{x}) = \exp(k_i \cdot \tilde{\rho}_i(\mathbf{x})) \quad (3)$$

where $\tilde{\rho}_i(\mathbf{x}) = \frac{\sum_{g=1}^G \rho_g M_g x_i^g}{C_i^{\text{cap}}}$ denotes the capacity-normalised offered load at UPF_i .

where ρ_g is the per-UE offered load for slice group g , M_g is the group population, and k_i is a node-specific exponent reflecting the processing capacity and queuing behaviour of UPF_i . The exponential form captures the onset of queuing congestion as utilisation approaches capacity. Since $k_{\text{mec}} \gg k_{\text{cc}}$, UPF_{mec} degrades faster under load than UPF_{cc} , reproducing the edge-core capacity asymmetry characteristic of stadium deployments [17]. The calibrated values $k_{\text{mec}} \approx 4.89$ and $k_{\text{cc}} \approx 0.49$ preserve the capacity ratio $k_{\text{mec}}/k_{\text{cc}} = 10$ per [3] while yielding realistic millisecond-range delays; empirical validation against a physical testbed is left for future work. The payoff for sessions in group g anchored at UPF_i is:

$$u_i^g(\mathbf{x}) = \frac{1}{d_i^g(\mathbf{x})} \quad (4)$$

A session is in latency violation when $d_i^g(\mathbf{x}) > \text{PDB}_g$. The instantaneous violation count L_t is used as the QoS indicator for the ML trigger in Section V.

F. EGT Replicator Dynamics

The SMF adapts session-to-UPF assignments by running the EGT replicator dynamics [3]. The review rate for sessions in group g at UPF_i is:

$$r_i^g(\mathbf{x}) = a - \beta u_i^g(\mathbf{x}), \quad \beta > 0, \quad \frac{a}{\beta} > u_i^g(\mathbf{x}) \quad (5)$$

Sessions with high payoff (low delay) are reviewed less frequently, stabilising their assignment. Population fractions evolve according to:

$$\dot{x}_i^g(t) = \beta \left(u_i^g(\mathbf{x}) - \bar{u}^g(\mathbf{x}) \right) x_i^g, \quad \bar{u}^g = \sum_i x_i^g u_i^g(\mathbf{x}) \quad (6)$$

The system converges to the evolutionary equilibrium $u_{\text{mec}}^g = u_{\text{cc}}^g$ for each group, at which all sessions achieve equal delay [3]. In the discrete-time implementation the SMF computes updated x_i^g values from Eq. (6) and issues N4/PFCP session modification rules redirecting newly arriving PDU sessions to the UPF indicated by the updated distribution. Established sessions are not re-anchored mid-flight within an iteration to avoid service interruption; the redistribution applies to new establishments and re-establishments only. The SMF-controllable UPF count N within a timing budget t_{service} is bounded by [3]:

$$N < F^{-1} \left(\frac{t_{\text{service}}}{L} \right) \quad (7)$$

where L is the number of iterations to convergence and F^{-1} is the inverse function relating per-iteration SMF processing delay to N .

subsectionEmulation Prototype

The methodology is validated on an OpenAirInterface (OAI) emulation testbed comprising OAI gNB as the 5G NR radio stack and OAI CN5G as the 5G standalone core, which natively supports multi-UPF deployment and network slicing via its NSSF-enabled configuration (AMF, SMF, UPF, NRF, UDM, AUSF, UDR, NSSF) [18]. Two OAI UPF instances are deployed as separate containerised processes, each with distinct GTP-U N3 tunnel endpoints and N6 interface bindings, registered to the NRF and associated with the SMF via PFCP over N4. Two network slices are provisioned with S-NSSAIs SST=1 (eMBB, 5QI 6) and SST=2 (URLLC, 5QI 82), with per-slice QoS flow parameters drawn from Table 5.7.4-1 of 3GPP TS 23.501 [11].

The EGT controller and BLSTM predictor are implemented in Python and interact with OAI CN5G via its Nsmf service-based interface (SBI) over HTTP/2, reading per-UPF session reports to compute $\mathbf{x}^g(t)$ and issuing updated N4/PFCP forwarding rules at each iteration. Traffic load traces emulate the three event phases of Section III-B: a 30-minute ramp-up from zero to $M_1 + M_2$ concurrent sessions, a 45-minute peak phase with a halftime demand spike modelled at $5.5 \times$ the pre-spike baseline (corresponding to the 450% increase reported in [14]), and a 20-minute dispersal phase.

IV. RESULTS AND DISCUSSION

Present quantitative and/or qualitative results using figures and tables. Discuss what the results mean, compare to baselines or state-of-the-art methods, and analyze strengths and limitations.

V. FUTURE WORK

A promising direction for future work is the integration of an ML-based reconfiguration trigger within the EGT control loop. While the current system relies on periodic scheduling, an intelligent trigger mechanism could enable event-driven reconfigurations, thereby reducing unnecessary overhead and improving responsiveness to rapid QoS degradations.

The envisioned trigger would leverage a Bidirectional LSTM (BLSTM) model trained on mobility-driven QoS degradation traces. Using three generic QoS indicators — the reconfiguration countdown T_g^{reconf} , the predicted session violation time T_g^{vio} , and the instantaneous violation count L_a — the model would predict impending violations and proactively initiate EGT re-convergence ahead of time.

This approach is expected to reduce reconfiguration overhead by at least 50% compared to periodic scheduling, while maintaining QoS compliance above 99.8% of simulation time. Validation on an OpenAirInterface (OAI) 5G NR emulation testbed using NSSF-enabled multi-UPF deployments is left as future experimental work.

VI. CONCLUSION

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